





CIRP - Requirements Gathering

Customer Intelligence & Retention Platform

DEPI • AI & Machine Learning Track • Round 5

Graduation Project

Project	Customer Churn Prediction & Segmentation Platform (CIRP)
Track	DEPI - AI & Machine Learning
Project Leader	Mohamed Atef Mohamed Abdo
Version / Date	v2.0 - November 2026
Repository	https://github.com/nhahub/NHA-5-011

1. Introduction

1.1 The project in short

CIRP is a small machine-learning project for an online shop. It reads the past orders and answers two questions: which customers are about to stop buying, and what kinds of customers do we have? The answers are shown in a simple web dashboard.

The project uses the public Online Retail II dataset. It has 22,372 customer records, 5,253 customers and five quarterly snapshots from September 2010 to September 2011. About 61% of the records are churn.

The 22 features come from the transaction invoices: invoice number and date, customer ID, product code and name, quantity and amount. Each feature uses only lines before the snapshot date.

1.2 Purpose of this document

This document says what the system must do. The design document uses it, and the final testing checks the system against it.

1.3 Scope

Table 1. Project scope

In scope	Out of scope
Cleaning the data and building quarterly customer records.	Sending e-mails or running campaigns.
Predicting churn in the next 90 days and explaining the score.	Scoring live website traffic in real time.
Grouping customers into six segments.	Names, addresses or payment data (the dataset has none).
A five-page dashboard and a CSV batch export.	A database server or user accounts.

2. Stakeholders

The people below use the system or check it. Each one needs a different level of detail.

Table 2. Stakeholders and their needs

Stakeholder	What they need	How they use CIRP
Marketing specialist	A short list of customers who need attention.	Opens one customer in the simulator, or scores a list on the batch page.
Retention manager	A ranked list to plan a campaign.	Uploads a CSV, filters by risk and segment, downloads the result.
Business management	The overall size of the churn problem.	Reads the KPI cards and the segment table on the overview page.
ML engineer	Models that can be rebuilt and checked.	Runs the training script and reads the evaluation report.
DEPI evaluator	Proof that the work is real and reproducible.	Reads the documents and follows the README steps.

3. User Stories

Table 3. User stories

ID	As a ...	I want to ...	So that ...	Priority
US-01	marketing specialist	see one customer's churn risk	I can decide whether to contact them	Must
US-02	marketing specialist	see why the model gave this score	I can write a relevant message	Must
US-03	marketing specialist	see the customer's segment	I use the right offer	Should
US-04	retention manager	score a whole CSV file	I do not open customers one by one	Must
US-05	retention manager	filter by risk and segment, then export	I can hand the list to the team	Must
US-06	business manager	see the main KPIs on one page	I can see the size of the problem	Must
US-07	ML engineer	have a saved model with its threshold	the dashboard gives the same result every time	Must
US-08	DEPI evaluator	know the exact steps to run the project	I can reproduce the numbers	Must

4. Use Cases

The first five use cases happen in the dashboard. The last three happen offline, when the team prepares the data or the models.

Table 4. Use case list

ID	Use case	Actor	Main steps	Result
UC-01	Score one customer	Marketing specialist	Pick a customer, edit the values, press score	Risk, level, top reasons, segment
UC-02	View segments	Marketing specialist	Open the segment page	Six cards with size and churn rate
UC-03	Batch scoring	Retention manager	Upload a CSV, check the columns, score	Table, KPIs, CSV download
UC-04	View overview	Business management	Open the overview page	KPIs and charts
UC-05	View model performance	ML engineer	Open the performance page	Metrics, curves, data dictionary
UC-06	Prepare the data	Data engineer	Run the data pipeline	22 features, labels, splits
UC-07	Train the model	ML engineer	Run the training script	Model file, threshold, metrics
UC-08	Compare deep learning	ML engineer	Run the deep-learning folder	Report and a keep-or-replace decision

4.1 Main flow of UC-01 and UC-03

1. The user opens the simulator page and picks a customer, or uploads a CSV on the batch page.
2. The application checks that all 22 features are present, numeric and in range.
3. The churn model returns a probability, and the stored threshold turns it into a label.
4. The segmentation model returns a cluster, and the profile table gives its name and action.
5. The page shows the risk level, the top three reasons and the segment, or offers the CSV download.

If a value is missing or the file has the wrong columns, the page explains the problem and does not score anything. If a model file is missing, the page says the model is not available.

5. Functional Requirements

All requirements below are for the first release. The letters in the ID show the module: D = data, P = prediction, S = segmentation, A = application, I = integration, ML = deep learning.

Table 5. Functional requirements

ID	Requirement	Priority
FR-D1	Read both sheets of the Excel workbook.	Must
FR-D2	Remove rows that appear in both sheets.	Must
FR-D3	Remove cancelled invoices, non-product codes and rows without a customer.	Must
FR-D4	Build five quarterly snapshots.	Must
FR-D5	Build the 22 features from data before the snapshot date.	Must
FR-D6	Label churn when the customer buys nothing in the next 90 days.	Must
FR-D7	Split by date: train, validation and test.	Must
FR-P1	Compare Logistic Regression, Decision Tree and Random Forest.	Must
FR-P2	Tune the Random Forest.	Must
FR-P3	Choose the threshold on the validation split only.	Must
FR-P4	Save the model, the threshold and the feature list in one file.	Must
FR-P5	Score the test split once, with the frozen model.	Must
FR-P6	Save the metrics and the evaluation figures.	Must
FR-P7	Keep the Random Forest as the production model.	Must
FR-S1	Compare K-Means and DBSCAN.	Must
FR-S2	Use six segments.	Must
FR-S3	Apply log1p and a scaler to the nine segmentation features.	Must
FR-S4	Give each segment a name, a description and one action.	Must
FR-S5	Show the size and the churn rate of each segment.	Should
FR-A1	Show four portfolio KPIs on the overview page.	Must
FR-A2	Show six segment cards with a short profile.	Must
FR-A3	Score one customer from a form.	Must
FR-A4	Turn the probability into a risk level.	Must
FR-A5	Show the top three reasons for the score.	Must
FR-A6	Score an uploaded CSV file.	Must
FR-A7	Check the 22 columns before scoring a file.	Must
FR-A8	Download the scored rows as a CSV.	Must
FR-I1	Use the same feature list offline and inside the app.	Must
FR-I2	Load saved models without retraining.	Must
FR-I3	Document how to install and run the project.	Must
FR-I4	Fix the random seed to 42.	Must
FR-ML1	Copy the dataset into an isolated folder.	Must
FR-ML2	Test at least 30 configurations of three networks.	Must
FR-ML3	Select the best network on validation only.	Must
FR-ML4	Evaluate the winner on the test split once.	Must
FR-ML5	Write a report with a keep-or-replace decision.	Must

6. Non-Functional Requirements

Table 6. Non-functional requirements

ID	Area	Requirement
NFR-1	Performance	One score in 2 s; 5,253 rows in 30 s; app start in 10 s.

ID	Area	Requirement
NFR-2	Quality	PR-AUC 0.80+ and recall 0.90+ on test; Silhouette 0.30+.
NFR-3	No leakage	Features use past data only. The test split is used once per model family.
NFR-4	Reproducibility	The same numbers, within 0.01, with seed 42 and the pinned library versions.
NFR-5	Privacy	Anonymised data only; uploaded files are not saved; no keys in the repository.
NFR-6	Usability	A new user finds a high-risk customer in 3 minutes or less.
NFR-7	Explainability	Reasons are written in plain words, not only as feature names.
NFR-8	Portability	Runs on Windows, macOS and Linux with relative paths.

7. Team and Weekly Tasks

7.1 Roles

Table 7. Roles and responsibilities

Member	Role	Main tasks	Tools
Mohamed Atef	Team lead, data and documents	Plan, data pipeline, documents, review	Python, pandas, Word, GitHub
Mohamed Ahmed	Preprocessing	Cleaning, duplicates, features, data dictionary	pandas, NumPy, openpyxl
Omar	Segmentation	K-Means, profiles, segment names	scikit-learn, matplotlib
Sohaib	Churn prediction	Models, threshold, evaluation, model file	scikit-learn, joblib
Mohamed Hamada	Deep learning	MLP, FT-Transformer and TabNet comparison, validation-only selection, report	PyTorch, pytorch-tabnet
Hussein	Integration and deep learning	Dashboard, batch page, input checks, tests; DL runs with Hamada	Streamlit, Plotly, PyTorch

7.2 Weekly plan

Table 8. Weekly task plan

Week	Dates	Main work	Owner
W1	12 - 18 Oct	Plan, dataset check, first cleaning rules	Ahmed, Atef
W2	19 - 25 Oct	Features, snapshots, churn label	Ahmed
W3	26 Oct - 1 Nov	Baseline models and first threshold	Sohaib
W4	2 - 8 Nov	Segments, profile table, segment page	Omar, Hussein
W5	9 - 15 Nov	Documents, dashboard pages, integration	Atef, Hussein
W6	16 - 22 Nov	Deep-learning search and one test run	Hamada, Hussein
W7	23 - 29 Nov	Performance test, fixes, final check	Hussein, all
W8	30 Nov - 4 Dec	Final report, presentation, GitHub upload	Atef, all

8. Constraints and Assumptions

- The dataset is public and anonymised (no names or payment data); there is no live shop connection, so the app reads saved CSV and model files.
- Everything runs on one laptop with no paid service or database server, and models are trained offline only.
- The 22 features are fixed, and the documents and code must be reproducible.